

Return Values: Getting Results from Your Functions

Most functions in trading systems need to provide results back to the caller. The `return` statement sends a value back from a function:

```
1. def calculate_sharpe_ratio(returns, risk_free_rate=0.02):
2.     # Calculate average return
3.     average_return = sum(returns) / len(returns)
4.
5.     # Calculate excess return (return above risk-free rate)
6.     excess_return = average_return - risk_free_rate
7.
8.     # Calculate standard deviation (volatility)
9.     squared_deviations = [(r - average_return) ** 2 for r in returns]
10.    variance = sum(squared_deviations) / len(returns)
11.    volatility = variance ** 0.5 # Square root of variance
12.
13.    # Calculate and return Sharpe ratio
14.    sharpe_ratio = excess_return / volatility
15.
16.    return sharpe_ratio
17.
```

This function calculates the Sharpe ratio—a measure of risk-adjusted returns that's essential in evaluating trading strategies—and returns the result.

Returning Multiple Values

Python functions can return multiple values at once, which is particularly useful for trading calculations that produce several related results:

```
1. def calculate_strategy_metrics(returns):
2.     # Calculate various performance metrics
3.     total_return = sum(returns)
4.
5.     # Calculate average return
6.     average_return = total_return / len(returns)
7.
8.     # Calculate volatility
```